



NeuroSpace

Complete evaluation export

Job ID	49b8cbe7-0f68-4997-955b-4fdb7ff3d328
Status	completed
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Pipeline Configuration

Parameter	Value
model_types	rnn, qrnn
comparison_mode	pca
feature_selector	pca
manual_overrides.rnn_epochs	30
manual_overrides.rnn_batch_size	64
manual_overrides.qrnn_epochs	15
manual_overrides.qrnn_batch_size	16
num_classes	6
class_names	1, 2, 3, 4, 5, 6
qrnn_backend	qiskit
noise_enabled	False
noise_level	0.0000
mitigation_enabled	False
mitigation_runs	1

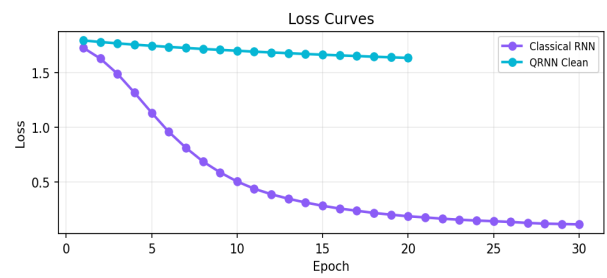
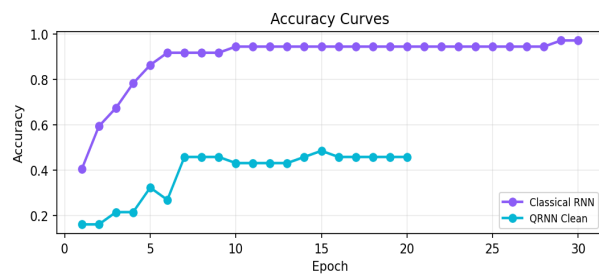
Best Model

Winner	RNN Accuracy	QRNN Accuracy	Improvement
RNN	93.24%	50.00%	43.24%

Evaluation Summary

Model	Accuracy	Precision	Recall	F1 Score	Loss
Classical RNN	93.24%	94.57%	93.24%	93.05%	0.1091
QRNN Clean	50.00%	48.44%	50.00%	46.97%	1.6373

Learning Curves



Classical RNN

Metric	Value
Accuracy	93.24%
Precision	94.57%
Recall	93.24%
F1 Score	93.05%
Loss	0.1091
Train Loss	0.1168
Val Loss	0.1091
Train Accuracy	98.82%
Val Accuracy	97.30%
Train Size	255
Val Size	37
Test Size	74

Classical RNN Confusion Matrix

Actual / Predicted	Class 0	Class 1				
Class 0	23	0	0	0	0	0
Class 1	0	8	0	2	1	1
Class 2	0	0	14	1	0	0
Class 3	0	0	0	10	0	0
Class 4	0	0	0	0	10	0
Class 5	0	0	0	0	0	4

Classical RNN Classification Report

Class	Precision	Recall	F1 Score	Support
1	100.00%	100.00%	100.00%	23.0000
macro avg	91.31%	93.33%	91.27%	74.0000
weighted avg	94.57%	93.24%	93.05%	74.0000
Accuracy			93.24%	

Classical RNN Epoch History

Epoch	Train Loss	Val Loss	Train Acc	Val Acc
1	1.9854	1.7304	18.43%	40.54%
2	1.6283	1.6309	34.51%	59.46%

Epoch	Train Loss	Val Loss	Train Acc	Val Acc
3	1.3099	1.4913	54.90%	67.57%
4	1.1368	1.3173	67.84%	78.38%
5	0.9347	1.1340	76.08%	86.49%
6	0.8764	0.9589	80.78%	91.89%
7	0.7575	0.8130	87.06%	91.89%
8	0.6275	0.6864	89.02%	91.89%
9	0.5731	0.5869	90.20%	91.89%
10	0.5175	0.5026	90.98%	94.59%
11	0.4593	0.4365	92.94%	94.59%
12	0.4311	0.3857	92.94%	94.59%
13	0.3569	0.3444	92.94%	94.59%
14	0.3379	0.3102	94.51%	94.59%
15	0.3294	0.2796	95.29%	94.59%
16	0.2780	0.2545	95.29%	94.59%
17	0.2563	0.2342	96.86%	94.59%
18	0.2607	0.2142	95.29%	94.59%
19	0.2279	0.1982	96.86%	94.59%
20	0.2167	0.1840	96.08%	94.59%
21	0.2085	0.1728	98.04%	94.59%
22	0.2346	0.1617	94.90%	94.59%
23	0.2018	0.1518	96.47%	94.59%
24	0.1793	0.1444	97.25%	94.59%
25	0.1653	0.1387	98.43%	94.59%
26	0.1484	0.1315	98.04%	94.59%
27	0.1524	0.1223	97.25%	94.59%
28	0.1391	0.1160	97.25%	94.59%
29	0.1349	0.1129	97.25%	97.30%
30	0.1168	0.1091	98.82%	97.30%

QRNN Clean

Metric	Value
Accuracy	50.00%
Precision	48.44%
Recall	50.00%
F1 Score	46.97%
Loss	1.6373
Train Loss	1.6195
Val Loss	1.6373
Train Accuracy	47.06%
Val Accuracy	45.95%
Train Size	255
Val Size	37
Test Size	74

QRNN Clean Confusion Matrix

Actual / Predicted	Class 0	Class 1				
Class 0	13	3	7	0	0	0
Class 1	0	6	0	5	0	1
Class 2	2	0	11	0	2	0
Class 3	0	3	0	7	0	0
Class 4	0	0	9	1	0	0
Class 5	0	3	0	1	0	0

QRNN Clean Classification Report

Class	Precision	Recall	F1 Score	Support
1	86.67%	56.52%	68.42%	23.0000
macro avg	36.23%	41.64%	37.26%	74.0000
weighted avg	48.44%	50.00%	46.97%	74.0000
Accuracy			50.00%	

QRNN Clean Epoch History

Epoch	Train Loss	Val Loss	Train Acc	Val Acc
1	1.8026	1.7974	9.41%	16.22%
2	1.7859	1.7837	9.80%	16.22%

Epoch	Train Loss	Val Loss	Train Acc	Val Acc
3	1.7699	1.7704	12.16%	21.62%
4	1.7543	1.7592	18.04%	21.62%
5	1.7400	1.7483	26.27%	32.43%
6	1.7306	1.7385	29.02%	27.03%
7	1.7192	1.7287	42.75%	45.95%
8	1.7111	1.7194	44.31%	45.95%
9	1.6977	1.7112	45.49%	45.95%
10	1.6914	1.7030	46.67%	43.24%
11	1.6833	1.6948	42.35%	43.24%
12	1.6739	1.6875	43.14%	43.24%
13	1.6632	1.6805	47.06%	43.24%
14	1.6599	1.6732	47.06%	45.95%
15	1.6500	1.6671	47.45%	48.65%
16	1.6435	1.6604	49.80%	45.95%
17	1.6370	1.6544	45.88%	45.95%
18	1.6294	1.6484	45.88%	45.95%
19	1.6242	1.6429	47.06%	45.95%
20	1.6195	1.6373	47.06%	45.95%